

1) what is javascript?

JS is a high level interpreted programming language mainly used to make web pages interactive and dynamic. It can be used for frontend development as well as backend development using technologies like Node.js.

Ex:

```
alert("Hello world");
```

2) what is variables;

A variable is a named container used to store data in a program. the stored value can be a number, string, boolean, object etc.

Ex:

```
let name = "priya";  
let age = 20;
```

name, age → variable

3) what is variable declaration?

variable declaration means creating a variable by specifying its name. In JS, we can declare variable using var, let or const.

Ex: let name: name is declared

4) what is initialization?

Initialization means giving an initial value to a variable when it is declared.

```
let age = 20;
```

let age → declaration  
= 20 → Initialization.



5) what is an assignment operator?

An assignment operator is used to assign a value to a variable. The most common assignment operator is `=`.

Ex: `let name;`  
`name = "Anbupriya";`

6) what are `var`, `let` and `const`;

`var`, `let` and `const` are keywords used to declare variables in javascript.

`var name = "Anbu";` older way of declaring variable

`let age = 20;` used when the value may change

`const country = "India";` used when the variable should not be reassigned.

8) what is ES6?

ES6 stands for ECMAScript 2015. It is a major version of the javascript language that introduced many new features to make js easier and more powerful.

important ES6 features.

\* `let` & `const` \* arrow function \* classes.

9) what is javascript engine?

A javascript is a program that reads, processes and executes javascript code. It converts javascript code into instructions that the computer can execute:

chrome  $\rightarrow$  V8

Firefox  $\rightarrow$  Spidermonkey.

Safari  $\rightarrow$  JavaScriptCore

Microsoft edge  $\rightarrow$  V8

10) what is V8?

V8 is a javascript and webassembly engine developed by Google. It is mainly written in C++ and is used by Google chrome and Node.js to execute js code.

7) what is the diff b/w let and const?

V8 Engine  
↳ Machine level instructions  
Execution

var,

| Feature        | var      | let          | const.       |
|----------------|----------|--------------|--------------|
| Scope          | Function | Block        | Block        |
| Re declaration | Allowed  | Not allowed  | Not allowed  |
| Reassignment   | Allowed  | Allowed      | Not allowed  |
| Hoisting.      | yes      | yes, but TDZ | yes, but TDZ |